EE 354/CE 361/MATH 310 -

Introduction to Probability and Statistics

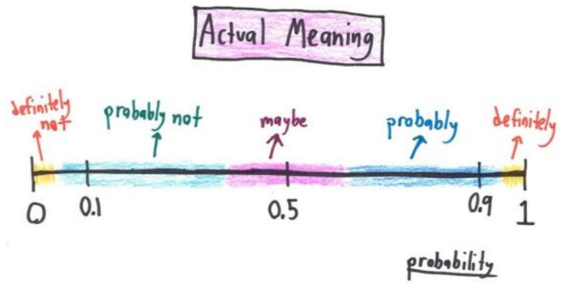
Spring 2026

EE 354/CE 361/MATH 310 L4 section (Spring 2026)

Introduction to Probability and Statistics -

Aamir Hasan

Week 14 - Lecture No 26 – April 13, 2026



Announcements!

1. No office hours on April 21, 2026
2. Grading update on Quiz No 07 –
    1. step 1– Marks out of 20, X = original Score
    2. Step 2 – $\frac{X}{16}$
    3. Step 3 – Marks in Step 2 scaled to 20

- Example 1 – if you scored 16, final score = 20.
- Example 2 – if you scored 12, Final Score = 15.

3. Planned quizzes
    1. Quiz No 08 – April 15, 2026 (Wednesday) – On Joint Distributions
    2. Quiz No 09 - April 22, 2026 (Wednesday)
    3. Quiz No 10 - April 29, 2026 (Wednesday)

Agenda for today

1. Unit 5 – Continuous Random Variable
    - ❑ Jointly Continuous Distributions

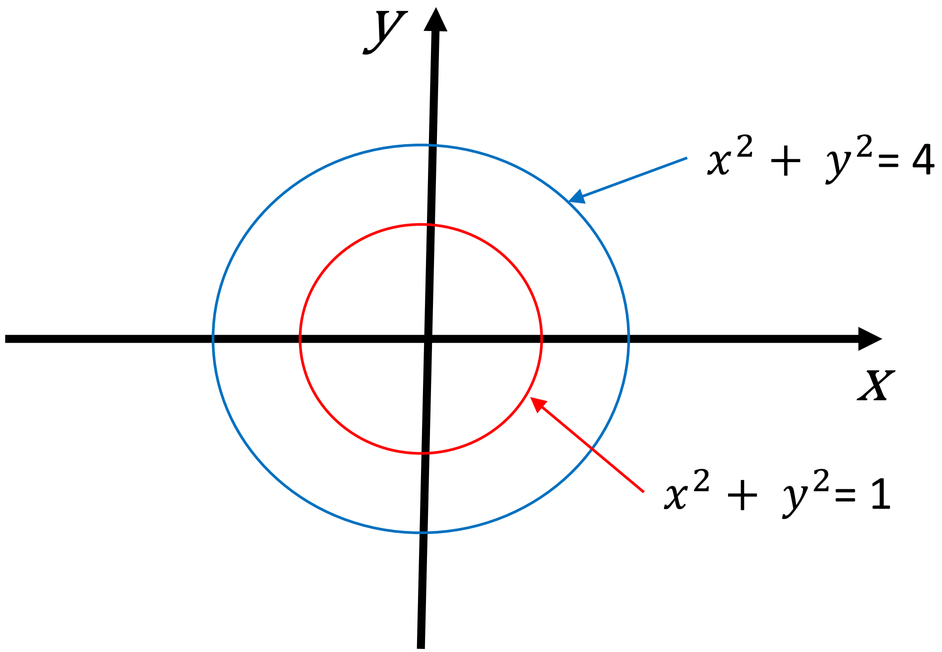
Independent Standard Random Variables

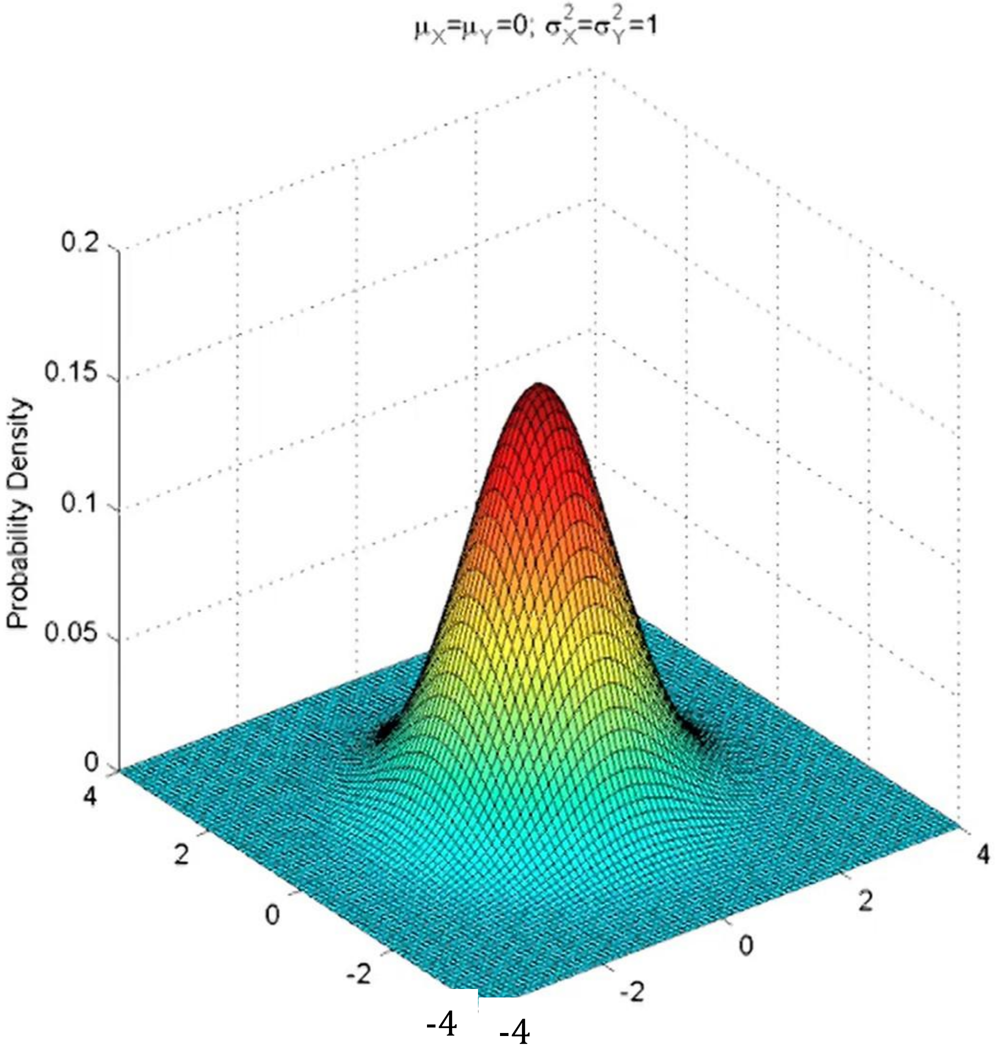
- General normal $N(\mu,\sigma^2)$: $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

$$f_{X,Y}(x,y) = f_X(x)\,.\,f_Y(y)$$

$$= \frac{1}{\sqrt{2\pi}}\exp\left\{-\frac{x^2}{2}\right\}.\frac{1}{\sqrt{2\pi}}\exp\left\{-\frac{y^2}{2}\right\}$$

$$= \frac{1}{2\pi}\exp\left\{-\frac{1}{2}(x^2+\ y^2)\right\}$$





Independent Standard Random Variables

$$f_{X,Y}(x,y) = f_X(x)\,.\,f_Y(y)$$

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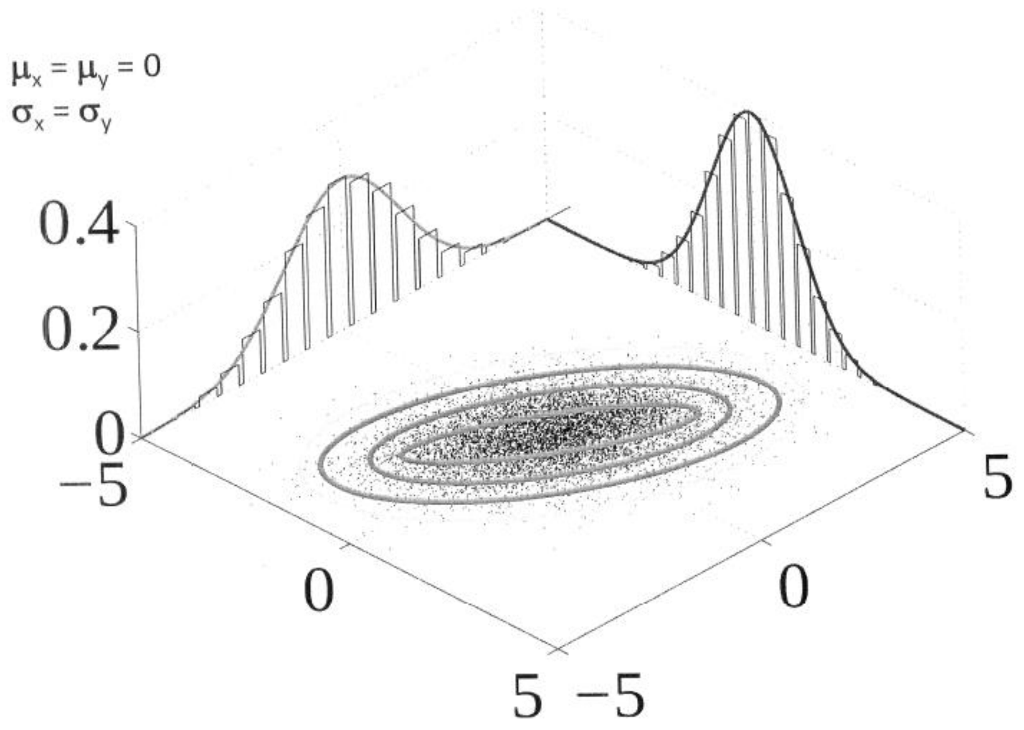
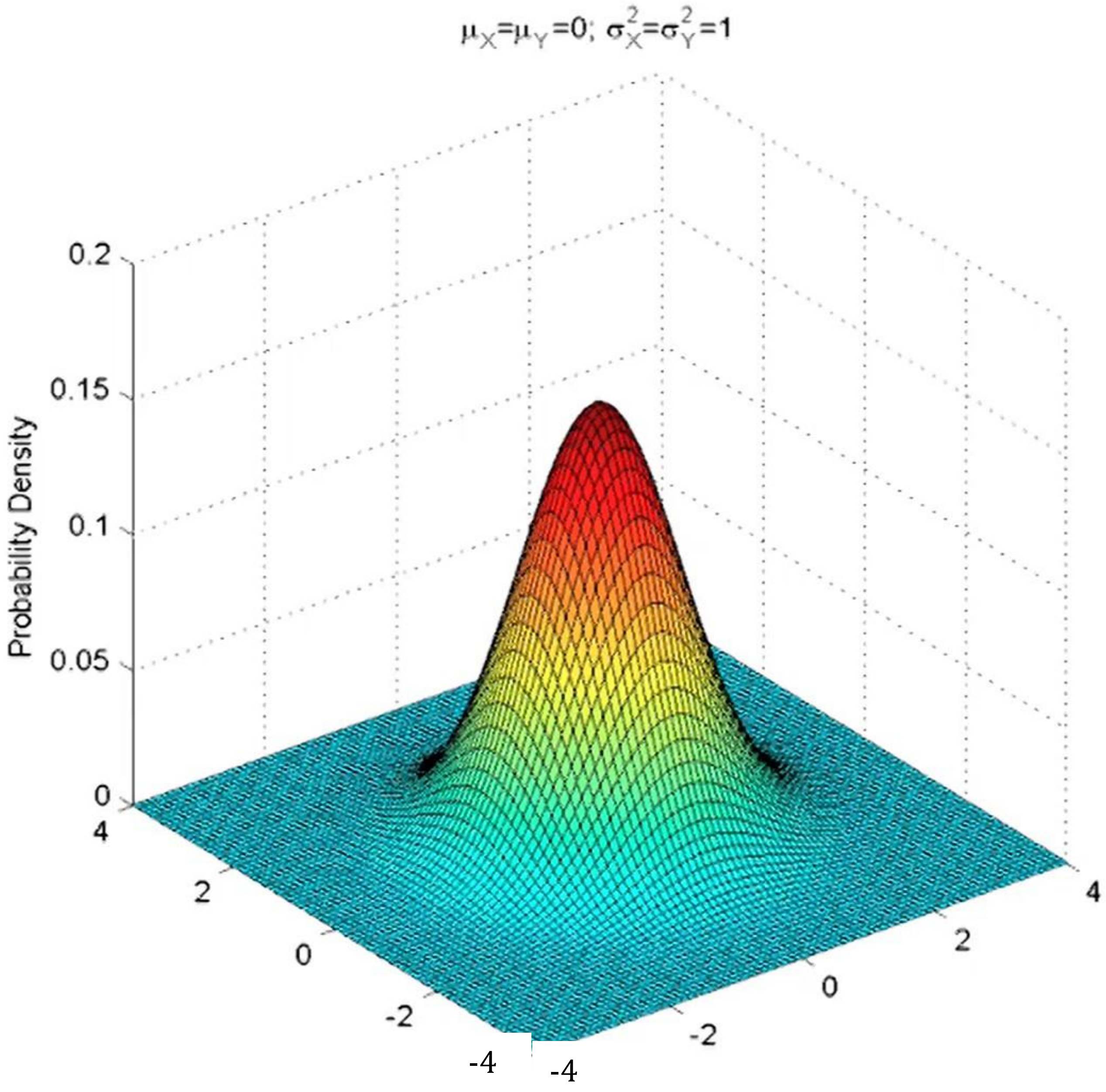


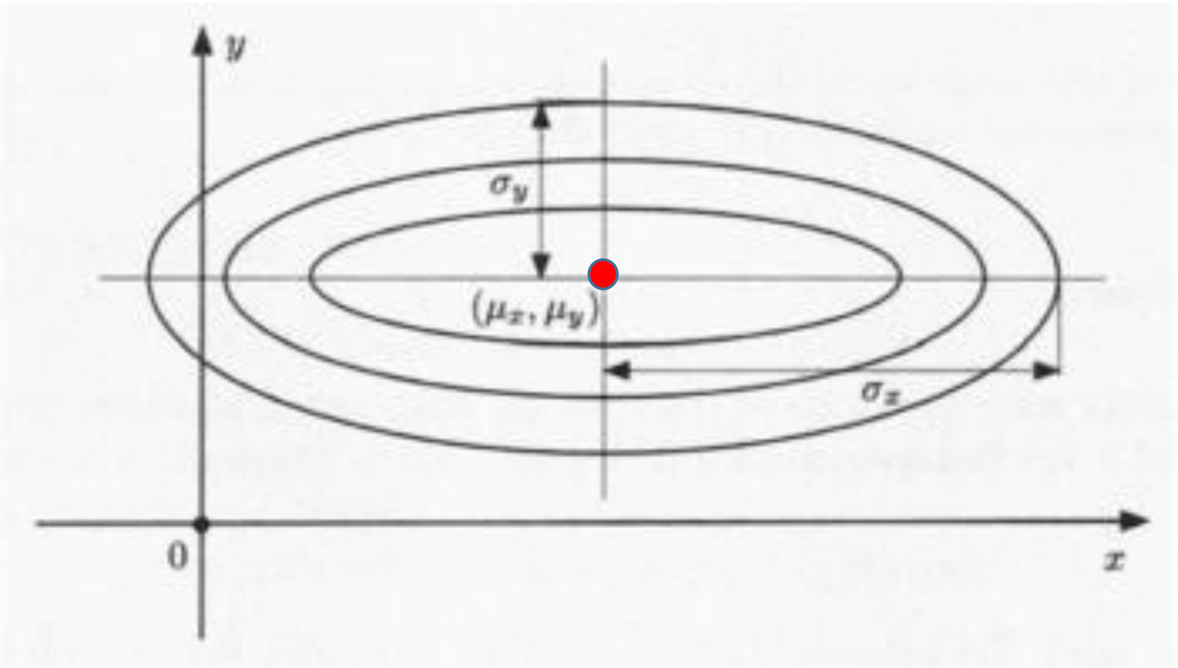


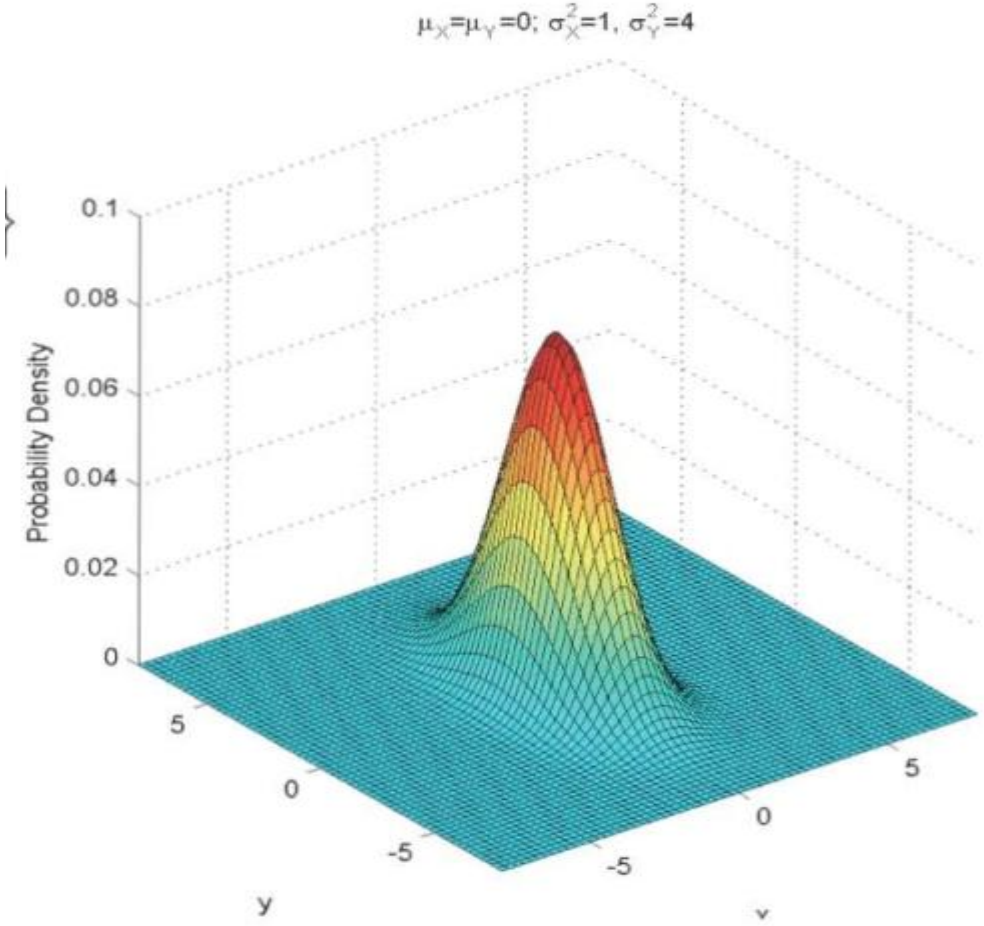
Independent Normal Random Variables

- General normal N(μ,σ^2): $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

$$f_{X,Y}(x,y) = f_X(x) f_Y(y)$$

$$= \underline{\frac{1}{2\pi\sigma_x\sigma_y}} \exp\left\{-\frac{(x-\mu_x)^2}{2\sigma_x{}^2} - \frac{(y-\mu_y)^2}{2\sigma_y{}^2}\right\}$$

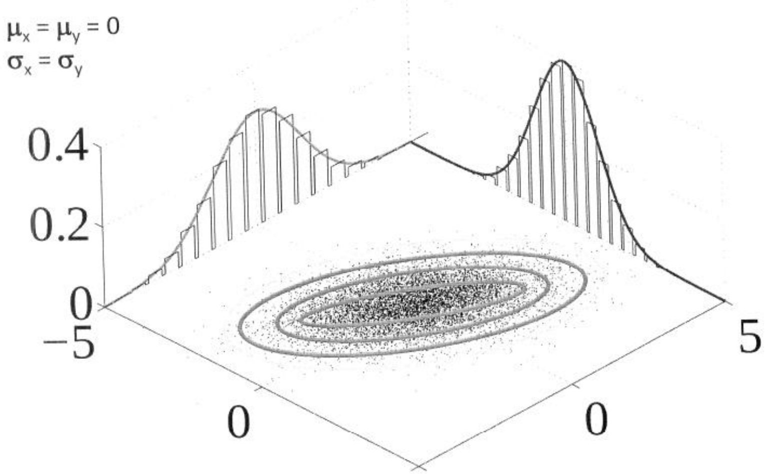




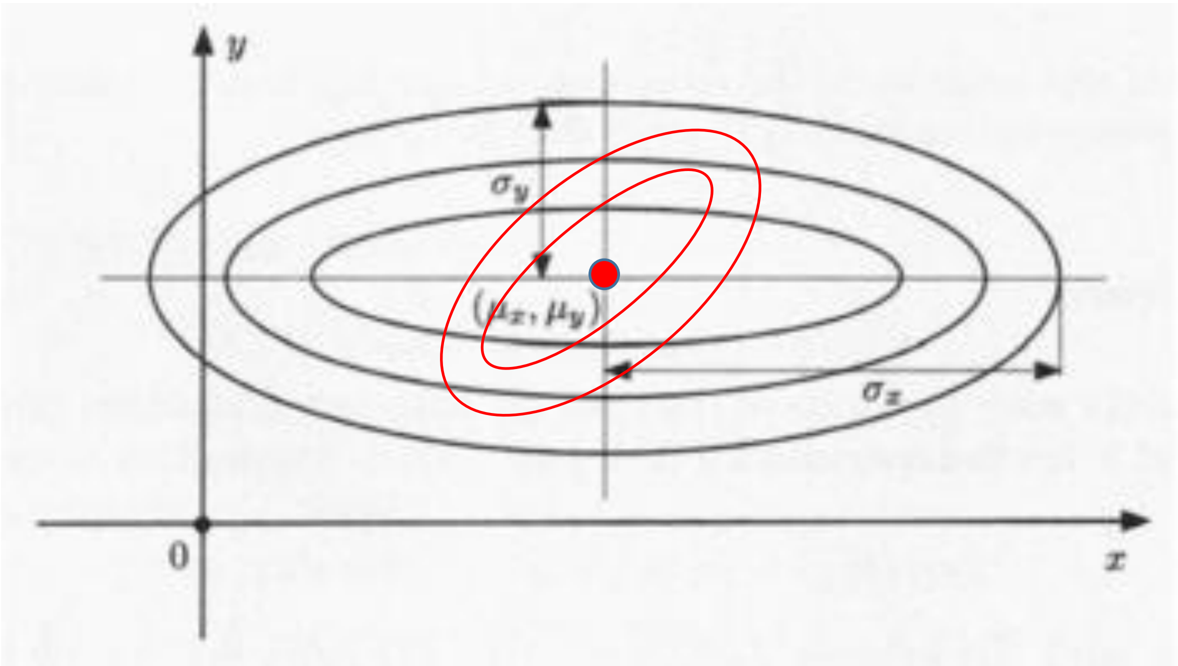
Bivariate Normal Random Variables

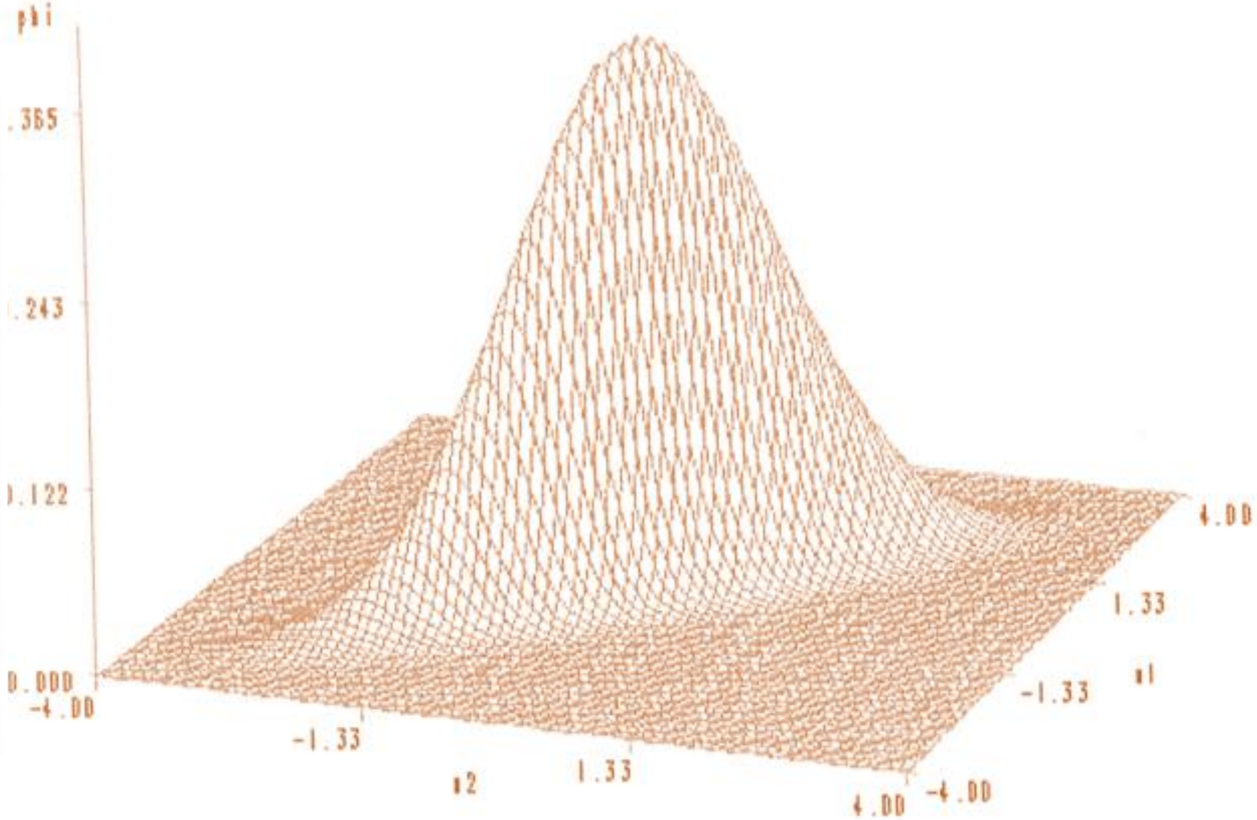
$$f_X(x) = \frac{1}{\sigma_x\sqrt{2\pi}} e^{-\frac{(x-\mu_x)^2}{2\sigma_x^{\ 2}}}$$

$$f_Y(y) = \frac{1}{\sigma_y\sqrt{2\pi}} e^{-\frac{(y-\mu_y)^2}{2\sigma_y^{\ 2}}}$$



$$f_{X,Y}(x,y)$$





Bivariate Normal Random Variables

Two random variables X and Y are said to be **bivariate normal**, or **jointly normal**, if a.X + b.Y has a normal distribution for all a, b ∈ R.

Example –

Let X~N(0,1) and W ~ Bernoulli (parameter = 1/2) be independent random variables.

Define the random variable Y: -

- Y = X, if W = 0 (w.p ½)
- Y = -X, if W = 1(w.p ½)

1. PDF of Y?
2. PDF of X + Y ?

$Y \sim N(0,1)$. Now, note that

$$Z = X + Y = \begin{cases} 2X & \text{with probability } \frac{1}{2} \\ 0 & \text{with probability } \frac{1}{2} \end{cases}$$

Mixed distribution (50% discrete, 50% Continuous)

From the joint to the marginals

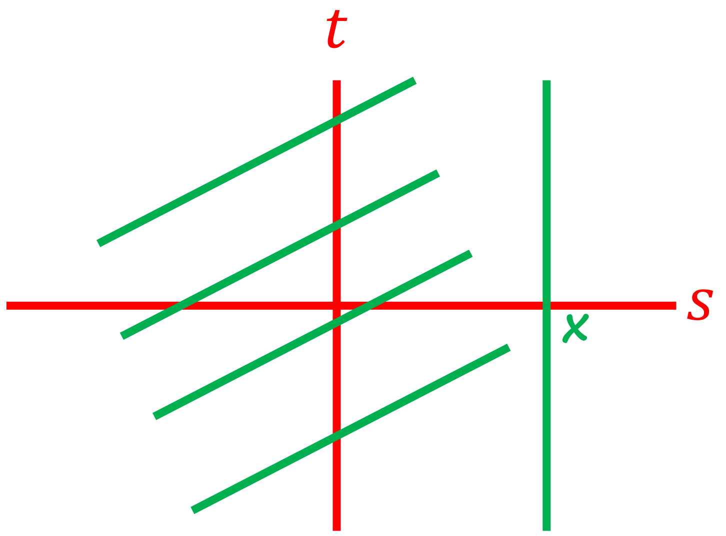
$$p_X(x) = \sum_y P_{X,Y}(x,y)$$

$$p_Y(y) = \sum_x P_{X,Y}(x,y)$$

$$f_X(x) = \int f_{X,Y}(x,y)\,dy$$

$$f_Y(y) = \int f_{X,Y}(x,y)\,dx$$

$$\Rightarrow \quad F_X(x) = P(X \le x) = \int_{-\infty}^{x} \left[\underline{\int_{-\infty}^{\infty} f_{X,Y}(s,t)\,dt} \right] ds$$

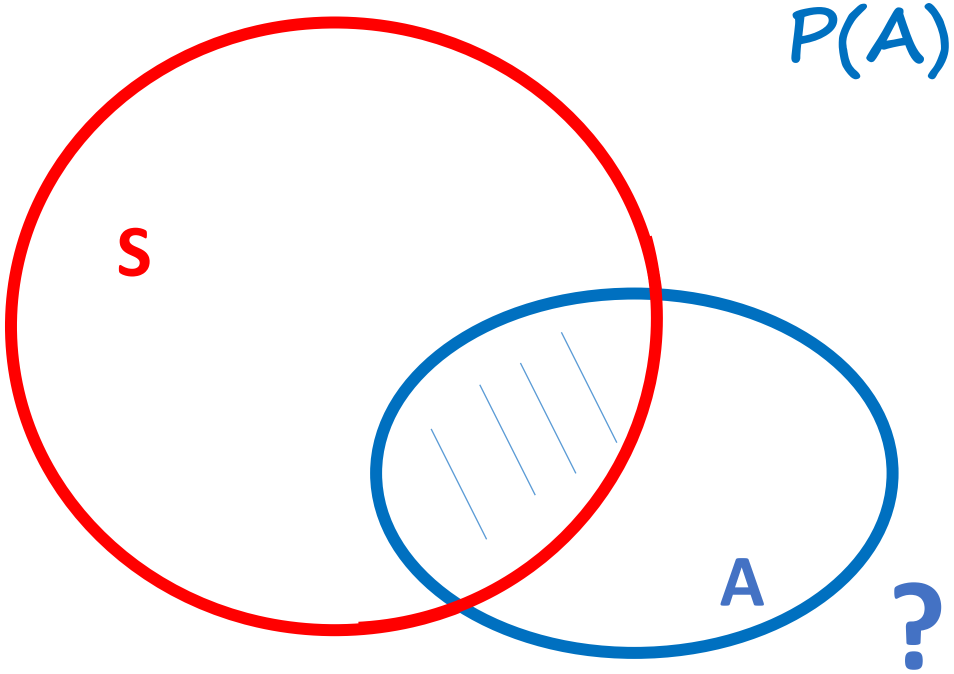


$$f_X(x) = \frac{dF_X}{dx}(x) = [\quad]$$

$$= \int_{-\infty}^{\infty} f_{X,Y}(s,t)\,dt$$

Uniform joint PDF on a set S

$$f_{X,Y}(x,y) = \begin{cases} \dfrac{1}{area\ of\ S}, & if\ (x,y)\ \in S, \\ 0, & Otherwise \end{cases}$$

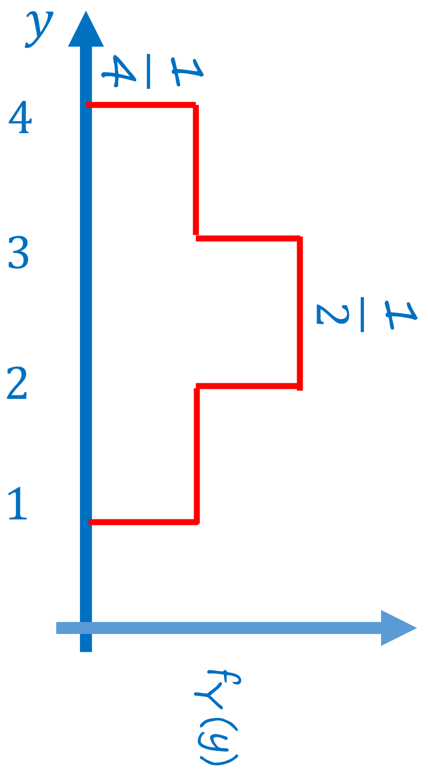


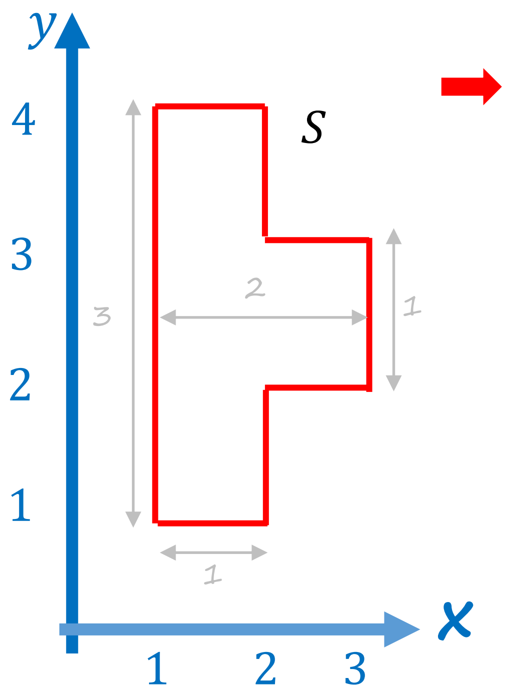
$$P(A) = \frac{area\ (A \cap S)}{area\ (S)}$$

Exercise - Uniform joint PDF on a set S

$f_X(x) = \int f_{X,Y}(x,y)\, dy$ $f_Y(y) = \int f_{X,Y}(x,y)\, dx$

$$f_{X,Y}(x,y) = \begin{cases} \dfrac{1}{area\ of\ S}, & if\ (x,y) \in S, \\ 0, & \text{Otherwise.} \end{cases}$$





➡ $f_{X,y}$ = ?

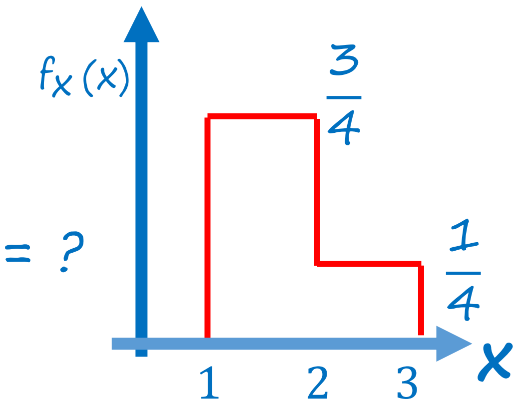
(e) $f_Y(y) = ?$

(a) $f_{X,y}(4, 4) = ?$

(b) $f_{X,y}(2, 2) = ?$

(c) $P(1 < X < 2, 1 < Y < 2) = ?$

(d) $f_X(x) = ?$



More than two random variables

- $p_{X,Y,Z}(x, y, z)$
- $\sum_{x}\sum_{y}\sum_{z} p_{X,Y,Z}(x, y, z) = 1$
- $p_X(x) = \sum_{z}\sum_{y} p_{X,Y,Z}(x, y, z)$
- $p_{X,Y}(x, y) = \sum_{z} p_{X,Y,Z}(x, y, z)$

- $f_{X,Y,Z}(x, y, z)$

Sum becomes integrals & pmfs becomes pdfs

Functions of multiple random variables

➡ $Z = g(X,Y)$

Expected value rule:

➡ $E[g(X,Y)] = \sum_x \sum_y g(x,y)\, p_{X,Y}(x,y)$ $\quad E[g(X,Y)] = \int\int g(x,y) f_{X,Y}(x,y)dxdy$

Linearity of expectations

➡ $E[aX+b] = aE[X]+b$

➡ $E[X+Y] = E[X]+E[Y]$

$$\boldsymbol{E[X_1 + \ldots + X_n] = E[X_1] + \ldots + E[X_n]}$$